



UNIVERSIDADE DE COIMBRA
FACULDADE DE CIÊNCIAS E TECNOLOGIA
Departamento de Engenharia Informática

Advanced Data Visualization (MECD)
Regular Season

09-06-2025

NAME: _____

Question Group 1 — General Concepts (20%)

- A)** Define “data”, “information” and “visualization” and explain how these terms are related in the context of data analysis.

- B)** What is the role of interaction in information visualization. Give an example of one advanced interaction technique.

- C) Define “expressiveness” and “effectiveness” principles in information visualization.

- D) Briefly describe the concept of **layout** in design of visualizations. Consider only visualization techniques/models, and not dashboards or other means of visual communication.

Question Group 2 — Visualization Analysis (20%)

- A) For each variable represented in the visualization (Figure 1) identify the following: (i) what **data attribute** is represented; (ii) its data type; (iii) what visual **mark** is used; (iv) and **channel** (visual property) is mapped to.

[illegible]

Figure 1

Data variable	Data type	Mark	Channel

Question Group 3 — Visualization Techniques (15%)

A) Tabular data representation: what is the main difference between stacked bar chart and polar area chart.

B) Multidimensional data: define SPLOM (Scatterplot Matrix) and sketch an example.

C) Geographic visualization: describe three alternative techniques used in flow maps

Question Group 4 — Critical Analysis (20%)

Considering the visualization in **Figure 2**, provide a critical analysis, listing all the identified problems.

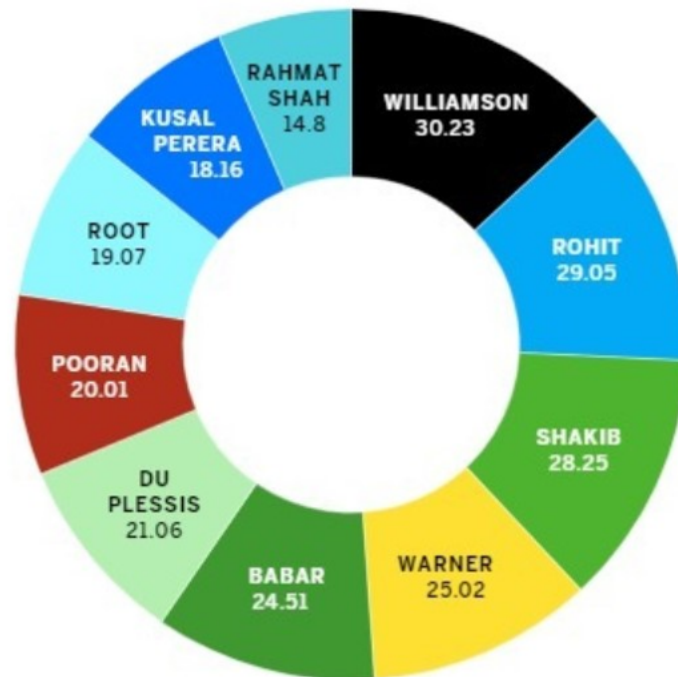


Figure 2

A) Provide a critical analysis.

B) Propose and sketch an alternative from of visualization, justifying the options taken.

Question Group 5 — 25%

A company that analyses banking transactions requires an analytical visualization to support analysts in their tasks. The main goal is to identify and analyse patterns of criminal activities. The analysts' tasks are simple: *i) identify cases of suspicious activities (e.g., abnormal amounts transferred in specific time period); ii) identify and analyse fraud schemes (e.g., money laundering).*

A typical pattern indicating money laundering can be characterized as follows: multiple transactions, generally made within a short time frame, to various accounts using the same device and involving one or multiple bank cards.

The dataset contains banking transactions of its clients. Each transaction is characterized by the following fields: *Client ID; Account ID; Timestamp; Device IP; Amount.*

You may assume that the analyst has access to various personal data of the clients available on the internet (e.g., social networks). You may also assume access to the full history of each client and that the dataset is complete (i.e., no missing data).

It is required to use **timeline-like** and **graph** representations. You may use any additional visualizations if necessary.

Using the knowledge and skills acquired in the data visualization course, propose an analytical visualization solution. Be creative in your response. You may propose innovative visualizations.

